

Circuit Documentation

1. Circuit Summary

This circuit is a two-wheel motor-control system built around an Arduino Nano, a TB6612FNG dual motor driver, two N20 DC gear motors, and a QTR-8A infrared reflectance sensor array. The circuit operates from a 12 V battery. A rocker switch controls the switched supply path. The switched battery voltage is connected to:

- The Arduino Nano VIN input
- The TB6612FNG motor supply input, VM
- The input of an LM2596 buck converter

The LM2596 module provides a regulated logic supply for the TB6612FNG VCC pin. The Arduino Nano supplies 5 V to the QTR-8A sensor array.

The Arduino reads the eight QTR-8A sensor outputs through analog inputs A0 through A7. It controls both channels of the TB6612FNG motor driver using digital direction, standby, and PWM signals.

No microcontroller code was supplied with the circuit, so the code section documents the required pin assignments and expected control functions rather than documenting an existing program.

Important electrical note: The LM2596 output must be adjusted and verified before connecting it to the TB6612FNG VCC pin. It should normally be set to approximately 5 V. The 12 V motor supply connected to VM must also be suitable for the specific N20 motors and within the TB6612FNG motor-supply rating.

2. Component List

Rocker Switch

A two-terminal power switch used to connect or disconnect the positive terminal of the 12 V battery from the rest of the circuit.

- Pin 1: Connected to the battery positive terminal
- Pin 2: Provides switched positive voltage to the circuit

N20 Motor (1)

The first DC gear motor, driven by motor channel A of the TB6612FNG motor driver.

- Motor terminals are connected to AO1 and AO2
- Direction depends on the polarity applied by the motor driver

QTR 8A

An eight-channel infrared reflectance sensor array. It provides eight sensor outputs that are read by the Arduino Nano.

- Powered from the Arduino Nano 5 V supply
- Ground is connected to the common circuit ground
- Sensor outputs are connected to analog inputs A0 through A7

The IR pin is not connected in the supplied net list.

N20 Motor

The second DC gear motor, driven by motor channel B of the TB6612FNG motor driver.

- Motor terminals are connected to BO1 and BO2
 - Direction depends on the polarity applied by the motor driver
-

Arduino Nano

The main microcontroller for the circuit. It reads the QTR-8A sensor array and generates control signals for both channels of the TB6612FNG motor driver.

It is powered through VIN from the switched battery supply and shares ground with the motor driver, sensor array, battery, and buck converter.

TB661 2FNG Motor Drive

A dual H-bridge motor driver used to independently control the two N20 motors.

- Channel A drives the first N20 motor
 - Channel B drives the second N20 motor
 - PWMA and PWMB provide speed control
 - AI1, AI2, BI1, and BI2 provide direction control
 - STBY enables or disables the driver
 - VM is the motor power input
 - VCC is the logic power input
-

12 V Battery

The primary energy source for the circuit.

- Positive terminal connects to the rocker switch
 - Negative terminal connects to the common ground
 - Supplies the motor driver and Arduino input through the switched power path
 - Also supplies the LM2596 input
-

Modul LM2596

A step-down switching regulator used to generate the logic supply for the TB6612FNG motor driver.

- Input is connected to the switched battery supply
- Output positive is connected to the TB6612FNG VCC
- Output negative is connected to circuit ground

The regulator output voltage should be adjusted to the required logic voltage before operation.

3. Component Wiring Details

Rocker Switch

- Pin 1 → 12 V battery positive
- Pin 2 → LM2596 In+
- Pin 2 → Arduino Nano VIN
- Pin 2 → TB6612FNG VM

The switch controls the positive supply to the complete powered circuit.

12 V Battery

- positive → Rocker switch pin 1
- negative → LM2596 In-
- negative → Arduino Nano GND
- negative → TB6612FNG GND

The battery negative terminal establishes the circuit reference ground.

Modul LM2596

- In+ → Rocker switch pin 2
- In- → 12 V battery negative
- In- → Arduino Nano GND
- In- → TB6612FNG GND
- Out+ → TB6612FNG VCC
- Out- → TB6612FNG GND

The LM2596 input receives switched battery voltage. Its output supplies the TB6612FNG logic rail.

The LM2596 Out- connection and the battery negative connection are part of the same ground system through the listed connections.

Arduino Nano

Power connections

- VIN → Rocker switch pin 2
- GND → 12 V battery negative
- GND → LM2596 In-
- GND → TB6612FNG GND
- 5V → QTR-8A VCC
- GND → QTR-8A GND

QTR-8A sensor connections

- A0 → QTR-8A D1
- A1 → QTR-8A D2
- A2 → QTR-8A D3
- A3 → QTR-8A D4
- A4 → QTR-8A D5
- A5 → QTR-8A D6
- A6 → QTR-8A D7
- A7 → QTR-8A D8

TB6612FNG control connections

- D5 → TB6612FNG PWMA
- D7 → TB6612FNG AI2
- D8 → TB6612FNG AI1
- D9 → TB6612FNG STBY
- D10 → TB6612FNG BI2
- D11 → TB6612FNG BI1
- D6 → TB6612FNG PWMB

The Arduino Nano pins D5 and D6 are PWM-capable on the standard Arduino Nano and are therefore suitable for motor speed control.

QTR 8A

- VCC → Arduino Nano 5V
- GND → Arduino Nano GND
- D1 → Arduino Nano A0
- D2 → Arduino Nano A1
- D3 → Arduino Nano A2
- D4 → Arduino Nano A3
- D5 → Arduino Nano A4
- D6 → Arduino Nano A5
- D7 → Arduino Nano A6
- D8 → Arduino Nano A7
- IR → Not connected

The sensor outputs are treated as analog inputs by the Arduino. The exact interpretation of the readings depends on the QTR-8A operating mode and the sensor hardware.

TB661 2FNG Motor Drive

Power connections

- VM → Rocker switch pin 2
- VCC → LM2596 Out+
- GND → 12 V battery negative
- GND → Arduino Nano GND
- GND → LM2596 In-
- GND → LM2596 Out-

Channel A control connections

- PWMA → Arduino Nano D5
- AI1 → Arduino Nano D8
- AI2 → Arduino Nano D7

Channel A motor connections

- AO1 → first N20 motor -VE
- AO2 → first N20 motor +VE

Channel B control connections

- PWMB → Arduino Nano D6
- BI1 → Arduino Nano D11
- BI2 → Arduino Nano D10

Channel B motor connections

- BO1 → second N20 motor -VE
- BO2 → second N20 motor +VE

Standby control

- STBY → Arduino Nano D9

The STBY input must be driven high by the Arduino for the motor driver to operate. Driving it low places the driver into standby.

The listed motor connections define the polarity used by the circuit. If a motor rotates opposite to the desired direction, its direction can be corrected in software by reversing the corresponding input logic or by exchanging its two motor leads.

N20 Motor (1)

- -VE → TB6612FNG AO1
- +VE → TB6612FNG AO2

This motor is controlled by channel A.

N20 Motor

- -VE → TB6612FNG BO1
- +VE → TB6612FNG BO2

This motor is controlled by channel B.

4. Electrical and Design Considerations

Supply-voltage arrangement

The switched battery voltage is connected directly to the Arduino Nano VIN pin and the TB6612FNG VM pin. This arrangement is only appropriate if:

- The battery voltage is within the Arduino Nano VIN input range.
- The battery voltage is within the permitted motor voltage range.
- The TB6612FNG motor supply is not exposed to a voltage above its maximum rating.
- The N20 motors are rated for the applied motor voltage.

The LM2596 regulator output is used only for the TB6612FNG logic supply in the supplied wiring.

Grounding

The Arduino Nano, sensor array, motor driver, battery, and LM2596 are connected to a common ground system. This common reference is required for reliable digital control and sensor readings.

Motor noise

DC motors can generate electrical noise and voltage transients. Motor-driver decoupling capacitors should be installed close to the TB6612FNG power pins. Additional suppression capacitors across the motor terminals may also be beneficial, provided they are suitable for the motor-driver application.

Unconnected sensor pin

The QTR-8A IR pin is not connected according to the supplied net list. If this pin controls the sensor emitter LEDs or another sensor function, the array may not operate as intended. The sensor documentation should be checked to determine whether the pin requires a supply, enable signal, or pull-up/down connection.

Unused TB6612FNG outputs

The listed circuit uses all four motor outputs:

- AO1 and AO2 drive the first motor.
- BO1 and BO2 drive the second motor.

No additional motor channels are unused.

5. Documented Code

No Arduino code was provided.

The following pin assignments should be used by the firmware:

```
// QTR-8A sensor inputs
const uint8_t sensorPins[8] = {
    A0, A1, A2, A3, A4, A5, A6, A7
};

// TB6612FNG channel A
const uint8_t PWMA = 5;
const uint8_t AI2  = 7;
const uint8_t AI1  = 8;

// TB6612FNG standby
const uint8_t STBY = 9;

// TB6612FNG channel B
const uint8_t BI2  = 10;
const uint8_t BI1  = 11;
const uint8_t PWMB = 6;
```

Required initialization

The firmware should configure:

- A0 through A7 as sensor inputs
- D5 and D6 as PWM outputs
- D7, D8, D10, and D11 as motor direction outputs
- D9 as the motor-driver standby output

The driver should initially be disabled by setting STBY low. After the system has initialized and valid sensor readings are available, STBY may be set high.

Example initialization structure:

```

void setup() {
  pinMode(PWMA, OUTPUT);
  pinMode(AI1, OUTPUT);
  pinMode(AI2, OUTPUT);

  pinMode(PWMB, OUTPUT);
  pinMode(BI1, OUTPUT);
  pinMode(BI2, OUTPUT);

  pinMode(STBY, OUTPUT);

  for (uint8_t i = 0; i < 8; i++) {
    pinMode(sensorPins[i], INPUT);
  }

  // Keep the motor driver disabled during startup
  digitalWrite(STBY, LOW);

  analogWrite(PWMA, 0);
  analogWrite(PWMB, 0);
}

```

Motor-control logic

For each motor channel:

- PWMA or PWMB controls speed using PWM.
- The two direction pins determine polarity.
- A PWM value of 0 commands zero drive.
- A PWM value of 255 commands maximum duty cycle.
- STBY must be high for the driver to operate.

A typical forward-control function could be structured as follows:

```

void setMotorA(int speed) {
  speed = constrain(speed, -255, 255);

  if (speed > 0) {
    digitalWrite(AI1, HIGH);
    digitalWrite(AI2, LOW);
    analogWrite(PWMA, speed);
  } else if (speed < 0) {
    digitalWrite(AI1, LOW);
    digitalWrite(AI2, HIGH);
    analogWrite(PWMA, -speed);
  } else {
    digitalWrite(AI1, LOW);
  }
}

```



```

        digitalWrite(AI2, LOW);
        analogWrite(PWMA, 0);
    }
}

void setMotorB(int speed) {
    speed = constrain(speed, -255, 255);

    if (speed > 0) {
        digitalWrite(BI1, HIGH);
        digitalWrite(BI2, LOW);
        analogWrite(PWMB, speed);
    } else if (speed < 0) {
        digitalWrite(BI1, LOW);
        digitalWrite(BI2, HIGH);
        analogWrite(PWMB, -speed);
    } else {
        digitalWrite(BI1, LOW);
        digitalWrite(BI2, LOW);
        analogWrite(PWMB, 0);
    }
}

```

The actual forward direction may differ depending on the physical motor orientation. If required, the logic for one motor can be inverted.

Sensor-reading structure

The eight sensor outputs can be read using:

```

int sensorValues[8];

void readSensors() {
    for (uint8_t i = 0; i < 8; i++) {
        sensorValues[i] = analogRead(sensorPins[i]);
    }
}

```

The sensor values can then be used for line detection, line-position estimation, closed-loop steering, or other autonomous control functions. Calibration and threshold values must be determined from the actual QTR-8A installation and surface conditions.